

Task 3

E/21/019

Assign all PCs to the same subnet (allocate appropriate IP addresses & Subnet masks) as in task 2.

1. Define 2 different VLANs (PC0 and PC1 in VLAN10, PC2 and PC3 in VLAN 20).

```
Switch1(config)#
Switch1#
00:13:00: %SYS-5-CONFIG_I: Configured from console by console
Switch1#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10, Fa0/11, Fa0/12 Fa0/13, Fa0/14, Fa0/15, Fa0/16 Fa0/17, Fa0/18, Fa0/19, Fa0/20 Fa0/21, Fa0/22, Fa0/23, Fa0/24 Gi0/1, Gi0/2
10	VLAN10	active	Fa0/1, Fa0/2
20	VLAN20	active	Fa0/3, Fa0/4
1002	fddi-default	act/unsup	
1003	token-ring-default	act/unsup	
1004	fddinet-default	act/unsup	
1005	trnet-default	act/unsup	

```
Switch1#
00:15:52: %LINK-3-UPDOWN: Interface FastEthernet0/1, changed state to up
00:15:53: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed stan
00:15:56: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed stan
00:15:57: %LINK-3-UPDOWN: Interface FastEthernet0/1, changed state to down
```

2. Observe the differences between the 2 settings (The same network with and without VLAN) using Wireshark.

Without VLAN

- All PCs are in **one broadcast domain**.
- ARP requests are broadcast to **all ports**.
- Any PC can communicate directly with any other PC.
- Wireshark shows:
 - ARP broadcasts are visible to all PCs
 - No VLAN tagging in Ethernet frames

```
C:\Users\hirus>ping 192.168.2.5
```

```
Pinging 192.168.2.5 with 32 bytes of data:  
Reply from 192.168.2.5: bytes=32 time=3ms TTL=128  
Reply from 192.168.2.5: bytes=32 time=1ms TTL=128  
Reply from 192.168.2.5: bytes=32 time=1ms TTL=128  
Reply from 192.168.2.5: bytes=32 time<1ms TTL=128
```

```
Ping statistics for 192.168.2.5:  
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),  
Approximate round trip times in milli-seconds:  
Minimum = 0ms, Maximum = 3ms, Average = 1ms
```

No.	Time	Source	Destination	Protocol	Length	Info
1	09:05:42.927532	Cisco_7f:26:07	Spanning-tree-for...	STP	60	Conf. Root = 32768/1/00:13:00:7f:26:00 Cost = 0 Port = 0x8007
2	09:05:44.115647	Cisco_7f:26:07	Cisco_7f:26:07	LOOP	60	Reply
3	09:05:44.927535	Cisco_7f:26:07	Spanning-tree-for...	STP	60	Conf. Root = 32768/1/00:13:00:7f:26:00 Cost = 0 Port = 0x8007
4	09:05:46.924146	192.168.2.3	192.168.2.5	ICMP	74	Echo (ping) request id=0x0001, seq=33/8448, ttl=128 (reply in 5)
5	09:05:46.925758	192.168.2.5	192.168.2.3	ICMP	74	Echo (ping) reply id=0x0001, seq=33/8448, ttl=128 (request in 4)
6	09:05:48.927560	Cisco_7f:26:07	Spanning-tree-for...	STP	60	Conf. Root = 32768/1/00:13:00:7f:26:00 Cost = 0 Port = 0x8007
7	09:05:47.935892	192.168.2.3	192.168.2.5	ICMP	74	Echo (ping) request id=0x0001, seq=34/8704, ttl=128 (reply in 8)
8	09:05:47.937645	192.168.2.5	192.168.2.3	ICMP	74	Echo (ping) reply id=0x0001, seq=34/8704, ttl=128 (request in 7)
9	09:05:48.927856	Cisco_7f:26:07	Spanning-tree-for...	STP	60	Conf. Root = 32768/1/00:13:00:7f:26:00 Cost = 0 Port = 0x8007
10	09:05:48.962165	192.168.2.3	192.168.2.5	ICMP	74	Echo (ping) request id=0x0001, seq=35/8960, ttl=128 (reply in 11)
11	09:05:48.963951	192.168.2.5	192.168.2.3	ICMP	74	Echo (ping) reply id=0x0001, seq=35/8960, ttl=128 (request in 10)
12	09:05:49.977289	192.168.2.3	192.168.2.5	ICMP	74	Echo (ping) request id=0x0001, seq=36/9216, ttl=128 (reply in 13)
13	09:05:49.978801	192.168.2.5	192.168.2.3	ICMP	74	Echo (ping) reply id=0x0001, seq=36/9216, ttl=128 (request in 12)
14	09:05:50.927666	Cisco_7f:26:07	Spanning-tree-for...	STP	60	Conf. Root = 32768/1/00:13:00:7f:26:00 Cost = 0 Port = 0x8007
15	09:05:51.738002	MicroStarINT_ac:7a:...	LCFCElectron_63:51:...	ARP	60	Who has 192.168.2.3? Tell 192.168.2.5
16	09:05:51.738028	LCFCElectron_63:51:...	MicroStarINT_ac:7a:...	ARP	42	192.168.2.3 is at 90:2e:16:63:51:f4
17	09:05:51.780764	LCFCElectron_63:51:...	MicroStarINT_ac:7a:...	ARP	42	Who has 192.168.2.5? Tell 192.168.2.3
18	09:05:51.782366	MicroStarINT_ac:7a:...	LCFCElectron_63:51:...	ARP	60	192.168.2.5 is at d8:bb:c1:ac:7a:cc
19	09:05:52.927264	Cisco_7f:26:07	Spanning-tree-for...	STP	60	Conf. Root = 32768/1/00:13:00:7f:26:00 Cost = 0 Port = 0x8007
20	09:05:54.115067	Cisco_7f:26:07	Cisco_7f:26:07	LOOP	60	Reply
21	09:05:54.927299	Cisco_7f:26:07	Spanning-tree-for...	STP	60	Conf. Root = 32768/1/00:13:00:7f:26:00 Cost = 0 Port = 0x8007
22	09:05:56.927296	Cisco_7f:26:07	Spanning-tree-for...	STP	60	Conf. Root = 32768/1/00:13:00:7f:26:00 Cost = 0 Port = 0x8007
23	09:05:58.937480	Cisco_7f:26:07	Spanning-tree-for...	STP	60	Conf. Root = 32768/1/00:13:00:7f:26:00 Cost = 0 Port = 0x8007

With VLAN

- VLAN 10 and VLAN 20 are **separate broadcast domains**.
- ARP broadcast from VLAN10 is **NOT received** by VLAN20.
- PCs in different VLANs cannot communicate without routing.
- Wireshark shows:
 - VLAN-tagged frames (802.1Q tag if trunk link used)
 - Broadcast is limited to its VLAN only

VLAN 10 - 192.168.2.4, 192.168.2.5

VLAN 20 - 192.168.2.2, 192.168.2.3

Same VLAN ping

```
C:\Users\Thimal Adeesha>ping 192.168.2.4
```

```
Pinging 192.168.2.4 with 32 bytes of data:
```

```
Reply from 192.168.2.4: bytes=32 time<1ms TTL=128
```

```
Reply from 192.168.2.4: bytes=32 time<1ms TTL=128
```

```
Reply from 192.168.2.4: bytes=32 time<1ms TTL=128
```

```
Reply from 192.168.2.4: bytes=32 time<1ms TTL=128
```

```
Ping statistics for 192.168.2.4:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
```

```
Approximate round trip times in milli-seconds:
```

```
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

```
C:\Users\hirus>ping 192.168.2.2
```

```
Pinging 192.168.2.2 with 32 bytes of data:
```

```
Reply from 192.168.2.2: bytes=32 time=1ms TTL=128
```

```
Reply from 192.168.2.2: bytes=32 time=1ms TTL=128
```

```
Reply from 192.168.2.2: bytes=32 time=1ms TTL=128
```

```
Reply from 192.168.2.2: bytes=32 time=1ms TTL=128
```

```
Ping statistics for 192.168.2.2:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
```

```
Approximate round trip times in milli-seconds:
```

```
    Minimum = 1ms, Maximum = 1ms, Average = 1ms
```

```
C:\Windows\System32>ping 192.168.2.3
```

```
Pinging 192.168.2.3 with 32 bytes of data:
```

```
Reply from 192.168.2.3: bytes=32 time=1ms TTL=128
```

```
Reply from 192.168.2.3: bytes=32 time<1ms TTL=128
```

```
Reply from 192.168.2.3: bytes=32 time<1ms TTL=128
```

```
Reply from 192.168.2.3: bytes=32 time=1ms TTL=128
```

```
Ping statistics for 192.168.2.3:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
```

```
Approximate round trip times in milli-seconds:
```

```
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

No.	Time	Source	Destination	Protocol	Length	Info
1	08:51:05.218367	Cisco 7f:26:03	Spanning-tree (for...	STP	60	Conf. Root = 32768/20/00:13:80:7f:26:00 Cost = 0 Port = 0x8003
2	08:51:07.218575	Cisco 7f:26:03	Spanning-tree (for...	STP	60	Conf. Root = 32768/20/00:13:80:7f:26:00 Cost = 0 Port = 0x8003
3	08:51:09.218817	Cisco 7f:26:03	Spanning-tree (for...	STP	60	Conf. Root = 32768/20/00:13:80:7f:26:00 Cost = 0 Port = 0x8003
4	08:51:11.218669	Cisco 7f:26:03	Spanning-tree (for...	STP	60	Conf. Root = 32768/20/00:13:80:7f:26:00 Cost = 0 Port = 0x8003
5	08:51:11.265257	Cisco 7f:26:03	CDP/VTP/DTP/PagP/UD...	CDP	386	Device ID: Switch1 Port ID: FastEthernet0/3
6	08:51:13.218608	Cisco 7f:26:03	Spanning-tree (for...	STP	60	Conf. Root = 32768/20/00:13:80:7f:26:00 Cost = 0 Port = 0x8003
7	08:51:14.087533	Cisco 7f:26:03	Cisco 7f:26:03	LOOP	60	Reply
8	08:51:15.218740	Cisco 7f:26:03	Spanning-tree (for...	STP	60	Conf. Root = 32768/20/00:13:80:7f:26:00 Cost = 0 Port = 0x8003
9	08:51:15.607582	192.168.2.2	192.168.2.3	ICMP	74	Echo (ping) request id=0x0001, seq=1/256, ttl=128 (reply in 10)
10	08:51:15.607568	192.168.2.3	192.168.2.2	ICMP	74	Echo (ping) reply id=0x0001, seq=1/256, ttl=128 (request in 9)
11	08:51:16.622078	192.168.2.2	192.168.2.3	ICMP	74	Echo (ping) request id=0x0001, seq=2/512, ttl=128 (reply in 12)
12	08:51:16.622267	192.168.2.3	192.168.2.2	ICMP	74	Echo (ping) reply id=0x0001, seq=2/512, ttl=128 (request in 11)
13	08:51:17.218752	Cisco 7f:26:03	Spanning-tree (for...	STP	60	Conf. Root = 32768/20/00:13:80:7f:26:00 Cost = 0 Port = 0x8003
14	08:51:17.633572	192.168.2.2	192.168.2.3	ICMP	74	Echo (ping) request id=0x0001, seq=3/768, ttl=128 (reply in 15)
15	08:51:17.633739	192.168.2.3	192.168.2.2	ICMP	74	Echo (ping) reply id=0x0001, seq=3/768, ttl=128 (request in 14)
16	08:51:18.651336	192.168.2.2	192.168.2.3	ICMP	74	Echo (ping) request id=0x0001, seq=4/1024, ttl=128 (reply in 17)
17	08:51:18.651499	192.168.2.3	192.168.2.2	ICMP	74	Echo (ping) reply id=0x0001, seq=4/1024, ttl=128 (request in 16)
18	08:51:19.218706	Cisco 7f:26:03	Spanning-tree (for...	STP	60	Conf. Root = 32768/20/00:13:80:7f:26:00 Cost = 0 Port = 0x8003
19	08:51:20.281838	LCFCElectron.63:51:...	MicroStarINT_a7:e6:...	ARP	42	Who has 192.168.2.2? Tell 192.168.2.3
20	08:51:20.283458	MicroStarINT_a7:e6:...	LCFCElectron.63:51:...	ARP	60	192.168.2.2 is at 04:7c:16:a7:e6:75
21	08:51:20.575644	MicroStarINT_a7:e6:...	LCFCElectron.63:51:...	ARP	60	Who has 192.168.2.3? Tell 192.168.2.2
22	08:51:20.575676	LCFCElectron.63:51:...	MicroStarINT_a7:e6:...	ARP	42	192.168.2.3 is at 90:2e:16:63:51:f4
23	08:51:21.218721	Cisco 7f:26:03	Spanning-tree (for...	STP	60	Conf. Root = 32768/20/00:13:80:7f:26:00 Cost = 0 Port = 0x8003
24	08:51:21.564917	192.168.2.2	192.168.2.3	ICMP	74	Echo (ping) request id=0x0001, seq=9/2304, ttl=128 (reply in 25)
25	08:51:21.565878	192.168.2.3	192.168.2.2	ICMP	74	Echo (ping) reply id=0x0001, seq=9/2304, ttl=128 (request in 24)
26	08:51:22.579910	192.168.2.2	192.168.2.3	ICMP	74	Echo (ping) request id=0x0001, seq=10/2560, ttl=128 (reply in 27)
27	08:51:22.580883	192.168.2.2	192.168.2.3	ICMP	74	Echo (ping) reply id=0x0001, seq=10/2560, ttl=128 (request in 26)
28	08:51:23.218399	Cisco 7f:26:03	Spanning-tree (for...	STP	60	Conf. Root = 32768/20/00:13:80:7f:26:00 Cost = 0 Port = 0x8003
29	08:51:23.607167	192.168.2.3	192.168.2.2	ICMP	74	Echo (ping) request id=0x0001, seq=11/2816, ttl=128 (reply in 30)
30	08:51:23.608268	192.168.2.2	192.168.2.3	ICMP	74	Echo (ping) reply id=0x0001, seq=11/2816, ttl=128 (request in 29)

Different VLAN ping

```
C:\Users\hirus>ping 192.168.2.4

Pinging 192.168.2.4 with 32 bytes of data:
Reply from 192.168.2.3: Destination host unreachable.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.2.4:
    Packets: Sent = 4, Received = 1, Lost = 3 (75% loss),
```

No.	Time	Source	Destination	Protocol	Length	Info
42	08:51:39.218972	Cisco 7f:26:03	Spanning-tree (for...	STP	60	Conf. Root = 32768/20/00:13:80:7f:26:00 Cost = 0 Port = 0x8003
43	08:51:41.219209	Cisco 7f:26:03	Spanning-tree (for...	STP	60	Conf. Root = 32768/20/00:13:80:7f:26:00 Cost = 0 Port = 0x8003
44	08:51:43.219086	Cisco 7f:26:03	Spanning-tree (for...	STP	60	Conf. Root = 32768/20/00:13:80:7f:26:00 Cost = 0 Port = 0x8003
45	08:51:44.087785	Cisco 7f:26:03	Cisco 7f:26:03	LOOP	60	Reply
46	08:51:45.223350	Cisco 7f:26:03	Spanning-tree (for...	STP	60	Conf. Root = 32768/20/00:13:80:7f:26:00 Cost = 0 Port = 0x8003
47	08:51:47.219128	Cisco 7f:26:03	Spanning-tree (for...	STP	60	Conf. Root = 32768/20/00:13:80:7f:26:00 Cost = 0 Port = 0x8003
48	08:51:49.219367	Cisco 7f:26:03	Spanning-tree (for...	STP	60	Conf. Root = 32768/20/00:13:80:7f:26:00 Cost = 0 Port = 0x8003
49	08:51:51.219593	Cisco 7f:26:03	Spanning-tree (for...	STP	60	Conf. Root = 32768/20/00:13:80:7f:26:00 Cost = 0 Port = 0x8003
50	08:51:53.219486	Cisco 7f:26:03	Spanning-tree (for...	STP	60	Conf. Root = 32768/20/00:13:80:7f:26:00 Cost = 0 Port = 0x8003
51	08:51:54.088051	Cisco 7f:26:03	Cisco 7f:26:03	LOOP	60	Reply
52	08:51:55.219364	Cisco 7f:26:03	Spanning-tree (for...	STP	60	Conf. Root = 32768/20/00:13:80:7f:26:00 Cost = 0 Port = 0x8003
53	08:51:55.298919	LCFCElectron.63:51:...	Broadcast	ARP	42	Who has 192.168.2.4? Tell 192.168.2.3
54	08:51:56.279488	LCFCElectron.63:51:...	Broadcast	ARP	42	Who has 192.168.2.4? Tell 192.168.2.3
55	08:51:57.219478	Cisco 7f:26:03	Spanning-tree (for...	STP	60	Conf. Root = 32768/20/00:13:80:7f:26:00 Cost = 0 Port = 0x8003
56	08:51:57.280214	LCFCElectron.63:51:...	Broadcast	ARP	42	Who has 192.168.2.4? Tell 192.168.2.3
57	08:51:59.219517	Cisco 7f:26:03	Spanning-tree (for...	STP	60	Conf. Root = 32768/20/00:13:80:7f:26:00 Cost = 0 Port = 0x8003
58	08:52:01.219624	Cisco 7f:26:03	Spanning-tree (for...	STP	60	Conf. Root = 32768/20/00:13:80:7f:26:00 Cost = 0 Port = 0x8003
59	08:52:03.219587	Cisco 7f:26:03	Spanning-tree (for...	STP	60	Conf. Root = 32768/20/00:13:80:7f:26:00 Cost = 0 Port = 0x8003
60	08:52:04.088272	Cisco 7f:26:03	Cisco 7f:26:03	LOOP	60	Reply
61	08:52:05.219658	Cisco 7f:26:03	Spanning-tree (for...	STP	60	Conf. Root = 32768/20/00:13:80:7f:26:00 Cost = 0 Port = 0x8003
62	08:52:07.219674	Cisco 7f:26:03	Spanning-tree (for...	STP	60	Conf. Root = 32768/20/00:13:80:7f:26:00 Cost = 0 Port = 0x8003
63	08:52:09.219707	Cisco 7f:26:03	Spanning-tree (for...	STP	60	Conf. Root = 32768/20/00:13:80:7f:26:00 Cost = 0 Port = 0x8003
64	08:52:11.219805	Cisco 7f:26:03	Spanning-tree (for...	STP	60	Conf. Root = 32768/20/00:13:80:7f:26:00 Cost = 0 Port = 0x8003
65	08:52:11.266406	Cisco 7f:26:03	CDP/VTP/DTP/PagP/UD...	CDP	386	Device ID: Switch1 Port ID: FastEthernet0/3
66	08:52:13.219592	Cisco 7f:26:03	Spanning-tree (for...	STP	60	Conf. Root = 32768/20/00:13:80:7f:26:00 Cost = 0 Port = 0x8003
67	08:52:14.088290	Cisco 7f:26:03	Cisco 7f:26:03	LOOP	60	Reply
68	08:52:15.219593	Cisco 7f:26:03	Spanning-tree (for...	STP	60	Conf. Root = 32768/20/00:13:80:7f:26:00 Cost = 0 Port = 0x8003
69	08:52:17.225480	Cisco 7f:26:03	Spanning-tree (for...	STP	60	Conf. Root = 32768/20/00:13:80:7f:26:00 Cost = 0 Port = 0x8003
70	08:52:19.223893	Cisco 7f:26:03	Spanning-tree (for...	STP	60	Conf. Root = 32768/20/00:13:80:7f:26:00 Cost = 0 Port = 0x8003
71	08:52:21.223973	Cisco 7f:26:03	Spanning-tree (for...	STP	60	Conf. Root = 32768/20/00:13:80:7f:26:00 Cost = 0 Port = 0x8003

```

C:\Users\Thimal Adeesha>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:
Reply from 192.168.2.5: Destination host unreachable.
Reply from 192.168.2.5: Destination host unreachable.
Reply from 192.168.2.5: Destination host unreachable.
Reply from 192.168.2.5: Destination host unreachable.

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

```

a. Discuss whether PCs in different VLANs should be assigned IP addresses in the same subnet.

- No

Reason:

- VLANs separate Layer 2 broadcast domains.
- Same subnet implies devices believe they are in same Layer 3 network.
- Communication will fail because Layer 2 isolation prevents ARP resolution.
- Causes confusion and misconfiguration.

3. Allocate PCs in separate VLANs with IPs in different subnets (e.g., PCs in VLAN10 given 192.168.10.0/24 and PCs in VLAN20 given 192.168.20.0/24), and observe the performances.

- VLAN10 → 192.168.10.0/24
- VLAN20 → 192.168.20.0/24

Same VLAN ping

```

C:\Users\hirus>ping 192.168.20.2

Pinging 192.168.20.2 with 32 bytes of data:
Reply from 192.168.20.2: bytes=32 time=3ms TTL=128
Reply from 192.168.20.2: bytes=32 time=1ms TTL=128
Reply from 192.168.20.2: bytes=32 time=1ms TTL=128
Reply from 192.168.20.2: bytes=32 time=1ms TTL=128

Ping statistics for 192.168.20.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 3ms, Average = 1ms

```

Different VLAN ping

```
C:\Users\hirus>ping 192.168.10.5
```

```
Pinging 192.168.10.5 with 32 bytes of data:
```

```
Request timed out.
```

```
Request timed out.
```

```
Request timed out.
```

```
Request timed out.
```

```
Ping statistics for 192.168.10.5:
```

```
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

No.	Time	Source	Destination	Protocol	Length	Info
11	09:09:33.748857	192.168.20.2	192.168.20.3	ICMP	74	Echo (ping) reply id=0x0001, seq=37/9472, ttl=128 (request in 10)
12	09:09:34.119027	Cisco 7f:26:03	Cisco 7f:26:03	LOOP	60	Reply
13	09:09:34.760494	192.168.20.3	192.168.20.2	ICMP	74	Echo (ping) request id=0x0001, seq=38/9728, ttl=128 (reply in 14)
14	09:09:34.761808	192.168.20.2	192.168.20.3	ICMP	74	Echo (ping) reply id=0x0001, seq=38/9728, ttl=128 (request in 13)
15	09:09:35.199330	Cisco 7f:26:03	Spanning-tree (for...	STP	60	Conf. TC + Root = 32768/20/00:13:00:7f:26:00 Cost = 0 Port = 0x8003
16	09:09:35.786313	192.168.20.3	192.168.20.2	ICMP	74	Echo (ping) request id=0x0001, seq=39/9984, ttl=128 (reply in 17)
17	09:09:35.787275	192.168.20.2	192.168.20.3	ICMP	74	Echo (ping) reply id=0x0001, seq=39/9984, ttl=128 (request in 16)
18	09:09:36.795452	192.168.20.3	192.168.20.2	ICMP	74	Echo (ping) request id=0x0001, seq=40/10240, ttl=128 (reply in 19)
19	09:09:36.796774	192.168.20.2	192.168.20.3	ICMP	74	Echo (ping) reply id=0x0001, seq=40/10240, ttl=128 (request in 18)
20	09:09:37.199632	Cisco 7f:26:03	Spanning-tree (for...	STP	60	Conf. TC + Root = 32768/20/00:13:00:7f:26:00 Cost = 0 Port = 0x8003
21	09:09:38.583888	MicroStarINT_a7:e6...	LCFCElectron_63:51...	ARP	60	Who has 192.168.20.3? Tell 192.168.20.2
22	09:09:38.583925	LCFCElectron_63:51...	MicroStarINT_a7:e6...	ARP	42	192.168.20.3 is at 90:2e:16:63:51:f4
23	09:09:39.198729	Cisco 7f:26:03	Spanning-tree (for...	STP	60	Conf. TC + Root = 32768/20/00:13:00:7f:26:00 Cost = 0 Port = 0x8003
24	09:09:41.198802	Cisco 7f:26:03	Spanning-tree (for...	STP	60	Conf. TC + Root = 32768/20/00:13:00:7f:26:00 Cost = 0 Port = 0x8003
25	09:09:42.996036	192.168.20.2	192.168.20.3	ICMP	74	Echo (ping) request id=0x0001, seq=9/2304, ttl=128 (reply in 26)
26	09:09:42.996219	192.168.20.3	192.168.20.2	ICMP	74	Echo (ping) reply id=0x0001, seq=9/2304, ttl=128 (request in 25)
27	09:09:43.199330	Cisco 7f:26:03	Spanning-tree (for...	STP	60	Conf. TC + Root = 32768/20/00:13:00:7f:26:00 Cost = 0 Port = 0x8003
28	09:09:44.027558	192.168.20.2	192.168.20.3	ICMP	74	Echo (ping) request id=0x0001, seq=10/2560, ttl=128 (reply in 29)
29	09:09:44.027752	192.168.20.3	192.168.20.2	ICMP	74	Echo (ping) reply id=0x0001, seq=10/2560, ttl=128 (request in 28)
30	09:09:44.119237	Cisco 7f:26:03	Cisco 7f:26:03	LOOP	60	Reply
31	09:09:44.722993	fe80::ad54:43aa:294...	ff02::1:2	DHCPv6	120	Information-request XID: 0xe8fc95 CID: 000100012a57d9a6902e166351f4
32	09:09:45.048292	192.168.20.2	192.168.20.3	ICMP	74	Echo (ping) request id=0x0001, seq=11/2816, ttl=128 (reply in 33)
33	09:09:45.048444	192.168.20.3	192.168.20.2	ICMP	74	Echo (ping) reply id=0x0001, seq=11/2816, ttl=128 (request in 32)
34	09:09:45.200851	Cisco 7f:26:03	Spanning-tree (for...	STP	60	Conf. TC + Root = 32768/20/00:13:00:7f:26:00 Cost = 0 Port = 0x8003
35	09:09:46.070517	192.168.20.2	192.168.20.3	ICMP	74	Echo (ping) request id=0x0001, seq=12/3072, ttl=128 (reply in 36)
36	09:09:46.070700	192.168.20.3	192.168.20.2	ICMP	74	Echo (ping) reply id=0x0001, seq=12/3072, ttl=128 (request in 35)
37	09:09:47.199342	Cisco 7f:26:03	Spanning-tree (for...	STP	60	Conf. TC + Root = 32768/20/00:13:00:7f:26:00 Cost = 0 Port = 0x8003
38	09:09:49.199398	Cisco 7f:26:03	Spanning-tree (for...	STP	60	Conf. TC + Root = 32768/20/00:13:00:7f:26:00 Cost = 0 Port = 0x8003
39	09:09:51.199417	Cisco 7f:26:03	Spanning-tree (for...	STP	60	Conf. TC + Root = 32768/20/00:13:00:7f:26:00 Cost = 0 Port = 0x8003

- PCs in the same VLAN communicate normally.
- PCs in different VLANs cannot communicate unless a router or Layer 3 switch is used.

a. Discuss the implications of assigning different subnets to different VLANs. Is this necessary?

- Improves organization
- Reduces broadcast traffic
- Improves security
- Easier routing configuration
- Standard network design practice

4. Connect two ports belonging to the two VLANs with a cable and observe the packet transfer.

- Directly connecting ports belonging to VLANs using a cable does not allow packet transfer because VLAN tags are not carried on access links. This behavior improves security by preventing VLAN hopping, broadcast leakage, unauthorized access, and man-in-the-middle attacks. Proper inter-VLAN communication must be done using trunk ports or Layer-3 routing devices.

```
Pinging 192.168.2.5 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.2.5:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

No.	Time	Source	Destination	Protocol	Length	Info
1	09:27:13.255160	Cisco_7f:26:03	Spanning-tree-(for...	STP	60	Conf. Root = 32768/20/00:13:80:7f:26:00 Cost = 0 Port = 0x8003
2	09:27:14.139785	Cisco_7f:26:03	Cisco_7f:26:03	LOOP	60	Reply
3	09:27:15.255190	Cisco_7f:26:03	Spanning-tree-(for...	STP	60	Conf. Root = 32768/20/00:13:80:7f:26:00 Cost = 0 Port = 0x8003
4	09:27:17.255421	Cisco_7f:26:03	Spanning-tree-(for...	STP	60	Conf. Root = 32768/20/00:13:80:7f:26:00 Cost = 0 Port = 0x8003
5	09:27:18.827939	LCFCElectron_63:51...	Broadcast	ARP	42	Who has 192.168.2.5? Tell 192.168.2.3
6	09:27:19.255276	Cisco_7f:26:03	Spanning-tree-(for...	STP	60	Conf. Root = 32768/20/00:13:80:7f:26:00 Cost = 0 Port = 0x8003
7	09:27:19.771120	LCFCElectron_63:51...	Broadcast	ARP	42	Who has 192.168.2.5? Tell 192.168.2.3
8	09:27:20.774798	LCFCElectron_63:51...	Broadcast	ARP	42	Who has 192.168.2.5? Tell 192.168.2.3
9	09:27:21.296572	Cisco_7f:26:03	Spanning-tree-(for...	STP	60	Conf. Root = 32768/20/00:13:80:7f:26:00 Cost = 0 Port = 0x8003
10	09:27:23.255339	Cisco_7f:26:03	Spanning-tree-(for...	STP	60	Conf. Root = 32768/20/00:13:80:7f:26:00 Cost = 0 Port = 0x8003
11	09:27:24.140012	Cisco_7f:26:03	Cisco_7f:26:03	LOOP	60	Reply
12	09:27:25.255382	Cisco_7f:26:03	Spanning-tree-(for...	STP	60	Conf. Root = 32768/20/00:13:80:7f:26:00 Cost = 0 Port = 0x8003
13	09:27:27.255456	Cisco_7f:26:03	Spanning-tree-(for...	STP	60	Conf. Root = 32768/20/00:13:80:7f:26:00 Cost = 0 Port = 0x8003

a. Discuss the security concerns of the above method.

- Bypasses VLAN isolation
- Allows unauthorized communication
- Can cause broadcast storms
- Enables VLAN hopping attacks
- Network instability

5. Allocate multiple (4 or more) PCs to a single VLAN. Allocate those devices with IP addresses in different subnets, with at least two PCs in a given subnet.

Subnet 1 - 192.168.1.0/24

Subnet 2 - 192.168.2.0/24

a. Observe the communication between devices.

Same subnet and different subnet ping

```
C:\Windows\System32>ping 192.168.1.5

Pinging 192.168.1.5 with 32 bytes of data:
Reply from 192.168.1.5: bytes=32 time<1ms TTL=128
Reply from 192.168.1.5: bytes=32 time<1ms TTL=128
Reply from 192.168.1.5: bytes=32 time<1ms TTL=128
Reply from 192.168.1.5: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.5:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Windows\System32>ping 192.168.2.3

Pinging 192.168.2.3 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.2.3:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Same subnet ping

```
Pinging 192.168.2.3 with 32 bytes of data:
Reply from 192.168.2.3: bytes=32 time<1ms TTL=128
Reply from 192.168.2.3: bytes=32 time<1ms TTL=128
Reply from 192.168.2.3: bytes=32 time<1ms TTL=128
Reply from 192.168.2.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.2.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

14	09:39:34.254247	192.168.2.3	192.168.2.5	ICMP	74 Echo (ping) reply	id=0x0001, seq=56/14336, ttl=128 (request in 13)
15	09:39:34.938741	Cisco 7f:26:07	Spanning-tree-for-...	STP	60 Conf. TC + Root = 32768/30/00:13:80:7f:26:00	Cost = 0 Port = 0x8007
16	09:39:35.772570	LCFCElectron_63:51:...	MicroStarINT_ac:7a:...	ARP	42 Who has 192.168.2.5? Tell 192.168.2.3	
17	09:39:35.774409	MicroStarINT_ac:7a:...	LCFCElectron_63:51:...	ARP	60 192.168.2.5 is at 08:bb:cl:ac:7a:cc	
18	09:39:36.938535	Cisco 7f:26:07	Spanning-tree-for-...	STP	60 Conf. TC + Root = 32768/30/00:13:80:7f:26:00	Cost = 0 Port = 0x8007
19	09:39:38.938714	Cisco 7f:26:07	Spanning-tree-for-...	STP	60 Conf. TC + Root = 32768/30/00:13:80:7f:26:00	Cost = 0 Port = 0x8007
20	09:39:39.794824	fe80::3df8:9854:32e...	ff02::1:2	DHCPv6	120 Information-request XID: 0xd8a5f CID: 00010001293ab553d8bbclac7acc	
21	09:39:40.938764	Cisco 7f:26:07	Spanning-tree-for-...	STP	60 Conf. TC + Root = 32768/30/00:13:80:7f:26:00	Cost = 0 Port = 0x8007
22	09:39:42.288297	fe80::ad54:43aa:294...	ff02::1:2	DHCPv6	120 Information-request XID: 0x4dfdl2 CID: 000100012a57d9a6902e166351f4	
23	09:39:42.542528	MicroStarINT_a7:e6:...	Broadcast	ARP	60 Who has 192.168.2.3? Tell 192.168.2.2	
24	09:39:42.542584	LCFCElectron_63:51:...	MicroStarINT_a7:e6:...	ARP	42 192.168.2.3 is at 90:2e:16:63:51:f4	
25	09:39:42.544854	192.168.2.2	192.168.2.3	ICMP	74 Echo (ping) request	id=0x0001, seq=13/3328, ttl=128 (reply in 26)
26	09:39:42.545015	192.168.2.3	192.168.2.2	ICMP	74 Echo (ping) reply	id=0x0001, seq=13/3328, ttl=128 (request in 25)
27	09:39:43.038532	Cisco 7f:26:07	Spanning-tree-for-...	STP	60 Conf. TC + Root = 32768/30/00:13:80:7f:26:00	Cost = 0 Port = 0x8007
28	09:39:43.552557	192.168.2.2	192.168.2.3	ICMP	74 Echo (ping) request	id=0x0001, seq=14/3584, ttl=128 (reply in 29)
29	09:39:43.552712	192.168.2.3	192.168.2.2	ICMP	74 Echo (ping) reply	id=0x0001, seq=14/3584, ttl=128 (request in 28)
30	09:39:44.158654	Cisco 7f:26:07	Cisco 7f:26:07	LOOP	60 Reply	
31	09:39:44.566051	192.168.2.2	192.168.2.3	ICMP	74 Echo (ping) request	id=0x0001, seq=15/3840, ttl=128 (reply in 32)
32	09:39:44.566218	192.168.2.3	192.168.2.2	ICMP	74 Echo (ping) reply	id=0x0001, seq=15/3840, ttl=128 (request in 31)
33	09:39:44.894968	192.168.2.5	224.0.0.251	MDNS	413 Standard query response 0x0000 TXT, cache flush PTR _ni-logos._tcp.local PTR MSI (5)._ni-logos._tcp.local SRV, cache flush 0 0 23...	
34	09:39:44.938715	Cisco 7f:26:07	Spanning-tree-for-...	STP	60 Conf. TC + Root = 32768/30/00:13:80:7f:26:00	Cost = 0 Port = 0x8007
35	09:39:45.578304	192.168.2.2	192.168.2.3	ICMP	74 Echo (ping) request	id=0x0001, seq=16/4096, ttl=128 (reply in 36)
36	09:39:45.578477	192.168.2.3	192.168.2.2	ICMP	74 Echo (ping) reply	id=0x0001, seq=16/4096, ttl=128 (request in 35)
37	09:39:46.938306	Cisco 7f:26:07	Spanning-tree-for-...	STP	60 Conf. TC + Root = 32768/30/00:13:80:7f:26:00	Cost = 0 Port = 0x8007
38	09:39:47.283730	LCFCElectron_63:51:...	MicroStarINT_a7:e6:...	ARP	42 Who has 192.168.2.2? Tell 192.168.2.3	
39	09:39:47.284318	MicroStarINT_a7:e6:...	LCFCElectron_63:51:...	ARP	60 192.168.2.2 is at 04:7c:16:a7:e6:75	
40	09:39:48.938404	Cisco 7f:26:07	Spanning-tree-for-...	STP	60 Conf. Root = 32768/30/00:13:80:7f:26:00	Cost = 0 Port = 0x8007
41	09:39:48.997892	192.168.2.3	224.0.0.251	MDNS	405 Standard query response 0x0000 TXT, cache flush PTR _ni-logos._tcp.local PTR HS._ni-logos._tcp.local SRV, cache flush 0 0 2343 HS...	
42	09:39:50.938404	Cisco 7f:26:07	Spanning-tree-for-...	STP	60 Conf. Root = 32768/30/00:13:80:7f:26:00	Cost = 0 Port = 0x8007

Different subnet ping

```
Pinging 192.168.1.4 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.1.4:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

b. Should devices in a given VLAN be allowed to assign any IP address they want?

- No.

Reasons:

- IP conflicts may occur.
- Wrong subnet assignment breaks communication.
- Security risk (IP spoofing).
- Network management becomes difficult.